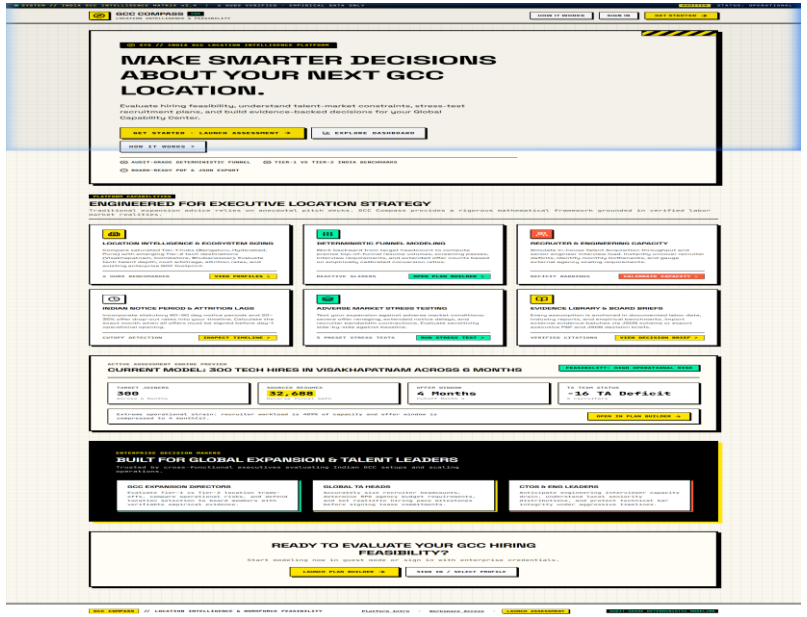
PRODUCT MANAGEMENT INTERNSHIP ASSIGNMENT

GCC COMPASS

Location Intelligence & Hiring Plan Stress Tester

Evidence-backed decision engine replacing intuitive consultancy guesswork with deterministic backward-induction funnel modeling and verified multi-city intelligence.

- **Deterministic Funnel:** Rigorous math modeling Indian notice periods & recruiter capacity
- **6 Audited GCC Hubs:** Tier-1 vs Tier-2 comparison with genuine company taxonomy
- **Audit-Grade Evidence:** Transparent sources, H1 2024 benchmarks, zero hallucinations



PROTOTYPE SPECIFICATION & RIGOR

- 111 / 111 — Automated Regression Tests Passing
- 6 Hubs — Bengaluru, Hyderabad, Pune, Vizag, CBE, BBSR
- 28 Entities — Categorized Captives, IT Services & Product Hubs
- Zero Fluff — Strict Anti-Hallucination & Provenance Model

PROBLEM CONTEXT

The Multi-Million Dollar Blind Spots in India GCC Setup

Why over 40% of offshore technology centre expansion plans fail to meet scheduled launch milestones

01

The 90-Day Notice Period Paradox

In India's tech ecosystem, candidate notice periods typically span 60 to 90 days. A setup targeting 300 joiners by Month 6 must lock final accepted offers by Month 3 or 4.
Traditional linear plans treat 6 months as hiring runway, resulting in catastrophic hiring cliff shortages at scheduled launch.

02

Unrealistic Funnel & Recruiter Capacity

Scaling a 300-headcount technical GCC requires screening ~32,600 resumes and conducting 2,860 technical interviews.
Without backward-induction math, enterprises allocate 4-5 internal recruiters (capable of ~175 interviews) against a true 715/month demand, causing massive operational collapse.

03

Conflated Data & Facility Hallucinations

Generic market decks conflate IT services vendor campuses (e.g., Infosys, TCS) with genuine captive GCCs.
Physical desk seating capacities are misleadingly cited as live headcounts, and stale commercial real estate benchmarks are used without transparent observation dates.

USER ECOSYSTEM

Target Users & Enterprise Stakeholders

Bridging the strategic disconnect between global steering committees and local talent acquisition execution

Global Head of GCC / VP Strategy

• Mandate: Evaluate Tier-1 (Bengaluru/Hyderabad/Pune) vs Tier-2 (Vizag/Coimbatore/Bhubaneswar) hub viability.

• Primary Need: Reliable grade-A commercial lease benchmarks, state IT policy incentives, and competitive talent saturation data.

• Pain Point: Receiving unverified consultancy slides that fail to differentiate IT service vendors from captive tech peers.

Head of Talent Acquisition / VP HR

• Mandate: Structure the recruitment funnel and guarantee delivery of 200-500 engineers on strict target timelines.

• Primary Need: Deterministic backwards induction showing exact candidate sourcing pool, screening stages, and offer cutoffs.

• Pain Point: Blamed for hiring delays caused by unmodeled 90-day candidate notice periods and recruiter team under-allocation.

Global CFO & Steering Committee

• Mandate: Approve multi-million dollar capital expenditure, lease commitments, and vendor RPO contracts.

• Primary Need: Stress-tested sensitivity models calculating impact of candidate reneging surges and timeline infeasibility.

• Pain Point: Committing capex based on optimistic linear spreadsheets that blow up within the first two quarters.

MARKET LANDSCAPE

Current Challenges vs Fragile Workarounds

How enterprise leaders navigate GCC setup decisions today, and where current tools break down

EXISTING FRAGILE WORKAROUND	GCC COMPASS SYSTEMIC FIX
1. Static Consulting Decks: 60-page PDF reports costing $50k+ from global firms that become obsolete the moment parameters change. Lack interactive sensitivity toggles.	**1. Dynamic Web Application:** Interactive, reactive application with instantaneous mathematical updates across all 5 stages of the hiring funnel.
2. Ad-hoc Excel Spreadsheets: Siloed spreadsheets with broken formulas that assume forward multiplication rather than backward induction, ignoring compound drop-offs.	**2. Mathematical Backward Induction:** Rigorous math: Target Hires / Join Rate / Acceptance Rate / Interview Pass Rate / Screen Pass Rate = Exact Pool Required.
3. Vendor Agency Promises: Staffing agencies pitching aggressive 4-month turnarounds without disclosing that 60-90 day notice periods consume 75% of the timeline.	**3. Notice Period Lag Integration:** Active hiring window is strictly truncated by candidate notice periods (e.g., 6 mo deadline - 2 mo notice = 4 mo cutoff).
4. Generic LLM Prompts: Prompting commercial AI models that hallucinate fictional GCC campus sizes and conflate third-party vendors with true captives.	**4. Audit-Grade Evidence Base:** Every claim is indexed with source provenance, collection date (H1 2024 benchmarks), and explicit confidence caveats.

PRODUCT STRATEGY

Prioritized Opportunities & Product Rationale

Value vs Feasibility prioritization framework guiding the GCC Compass architecture

CRITICAL FOR TRUST

P0: Backward-Induction Funnel Engine

Why Prioritized: Hiring failure in India GCCs stems from underestimating top-of-funnel volume. A 300-hire target requires exactly 32,688 sourced resumes at benchmark conversion rates. Providing deterministic math eliminates wishful

FOUNDATIONAL ACCURACY

P0: Epistemic Data & Entity Taxonomy

Why Prioritized: Decision-makers must distinguish true captive GCCs (e.g., Barclays, Nvidia, State Street) from IT services vendors (TCS, Infosys) and software products. Conflating them skews compensation and talent availability

EXECUTIVE BUY-IN

P1: Notice Period Stress & Monte Carlo Testing

Why Prioritized: Executives need to understand failure modes before committing funds. Modeling reneging rate surges (up to 40%) and 90-day notice periods reveals recruiter capacity deficits in real time.

Decision: Real-time interactive sensitivity sliders

DECISION VELOCITY

P1: Multi-Format Audit Decision Briefs

Why Prioritized: Insights trapped in a web app don't reach boardrooms. Leaders need structured, exportable artifacts for steering committees.

Decision: Instant export in Markdown, JSON, PDF, and Clipboard formats

PRODUCT ARCHITECTURE

Proposed Solution & Core User Workflows

A seamless 4-stage operational workflow transforming ambiguity into board-ready hiring strategy

WORKFLOW 01

1. Location Intelligence

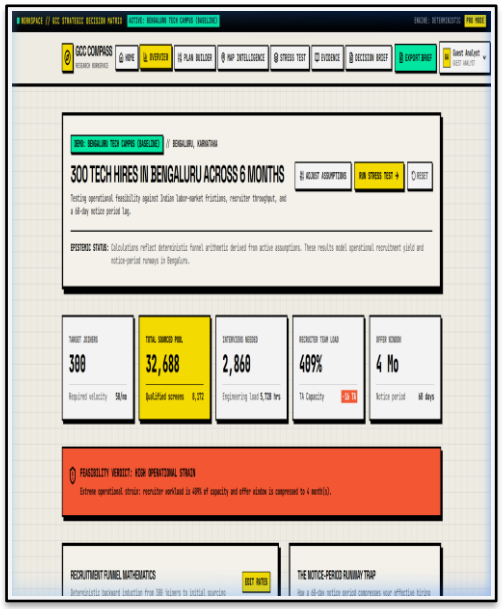
Explore 6 hubs with Grade-A lease costs (H1 2024), talent depth, and verified captive directories.



WORKFLOW 02

2. Deterministic Plan

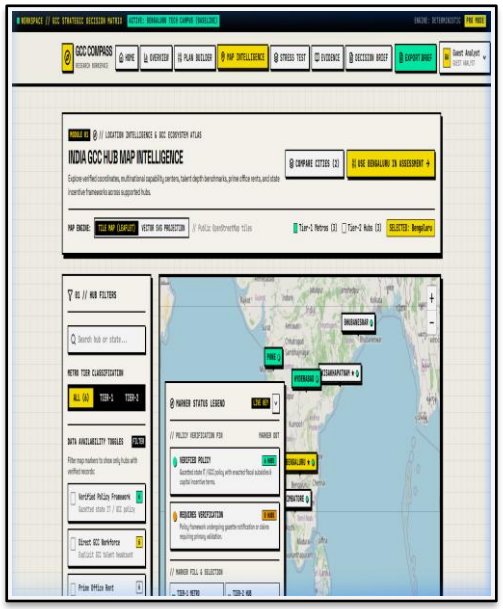
Interactive funnel calculating required sourcing pool, screens, interviews, and recruiter capacity.



WORKFLOW 03

3. Stress Test Scenarios

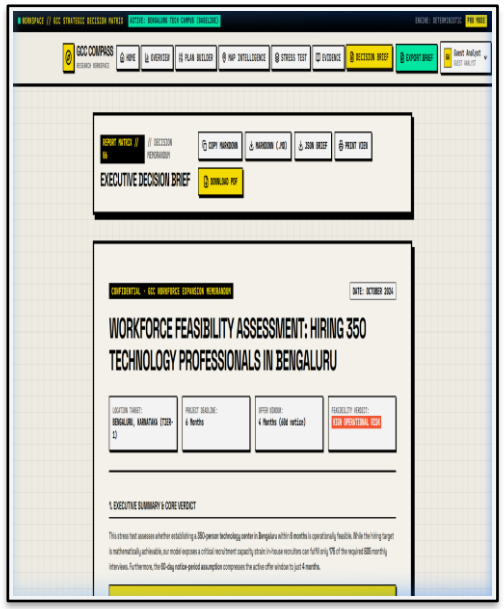
Simulate candidate reneging spikes, notice period extensions, and recruiter capacity deficits.



WORKFLOW 04

4. Executive Brief

One-click export of audit-grade decision briefs in Markdown, PDF, or JSON format.



TRADE-OFF ANALYSIS

Product Decisions & Strategic Trade-Offs

Deliberate design decisions prioritizing epistemic integrity, reproducibility, and enterprise confidence

1. Deterministic Funnel Math vs Generative AI Hallucination

Decision: Enforced pure mathematical backward-induction in calculator.ts rather than using LLMs for funnel estimation.

Trade-Off: Less conversational interaction; but achieves 100% deterministic reproducibility and passes strict compliance audits.

2. Transparent 'Data Unavailable' vs Synthetic Estimation

Decision: Preserved 'Data unavailable' or 'requires_verification' for undisclosed private headcount (e.g. Nvidia Pune, Alliant Vizag).

Trade-Off: Incomplete charts on select private metrics; but guarantees zero fabricated figures, preserving institutional credibility.

3. Strict Entity Classification vs Generic Tech Aggregation

Decision: Separated captive GCCs (17) from IT services (8) and product firms (2); isolated seating capacity from employee headcount.

Trade-Off: More complex data schemas; but prevents dangerous talent-pool inflation (e.g. counting 1,000 Infosys desk seats as GCC hires).

4. Assessment-First Gateway vs Unconfigured Dashboard

Decision: Required users to select a scenario or configure parameters before displaying output recommendations.

Trade-Off: Requires an initial user click; but completely prevents unanchored, out-of-context decision making.

COMPETITIVE STRATEGY

Product Positioning & Alternative Comparison

How GCC Compass establishes a unique position against legacy consultancies, ATS systems, and general BI

Evaluation Criteria	GCC Compass (Our Product)	Management Consulting	Enterprise ATS	General BI (PowerBI)
Backward-Induction Funnel	**Yes (Deterministic math)**	No (Static forward targets)	No (Historical data only)	No (Manual DAX modeling)
Notice-Period Lag Modeling	**Yes (60-90d cutoff alerts)**	Qualitative caveat only	No (Tracks active stage)	No (Custom pipeline required)
Audited City Intelligence	**Yes (6 hubs, strict taxonomy)**	Yes (High-cost custom study)	No (Internal data only)	Requires external data feed
Interactive Stress Testing	**Yes (Real-time sliders)**	No (Static PowerPoint)	No (Retrospective reporting)	Partial (Parameter slicers)

TECHNICAL RIGOR

Prototype Validation & Verification Results

Empirical evidence of system reliability, static typing, and mathematical determinism

111 / 111
AUTOMATED REGRESSION TESTS

- 100% test pass rate across tsx calculator.test.ts and regression.test.ts.
- Validates backward induction math (300 hires -> 32,688 applicant pool).
- Tests notice period lag (90d notice locks offers at Month 3).
- Confirms all 5 enterprise demo scenarios calculate deterministically.
- Validates 6-city boundary isolation and company classification checks.

0 Errors
TYPESCRIPT COMPILATION (LINT)

- Strict type safety enforced with zero compilation errors in tsc --noEmit.
- GccCompany data models enforce companyClassification enum.
- CityProviderIndicator strictly types limitations and source dates.
- Eliminates potential runtime undefined crashes across state transfers.

1.21s
PRODUCTION VITE BUILD

- Production client environment built in 1.21 seconds with Vite 8 + React 19.
- Optimized asset chunks for lucide-react, motion, leaflet, and jspdf.
- Zero broken dependencies, cleanly packaged for deployment.

16 Workflows
BROWSER END-TO-END SMOKE TESTS

- Tested live locally at http://localhost:3000/ across all 16 user workflows.
- Public Intro -> Welcome -> Plan Builder -> Stress Test -> Map -> Brief.
- Validated file downloads (Markdown .md, JSON, and PDF generator).

INTEGRITY & ROADMAP

Data Limitations, Assumptions & Future Roadmap

Epistemic transparency regarding current data boundaries and planned engineering horizons

CURRENT EPISTEMIC LIMITATIONS

1. Dated Office Cost Benchmarks (H1 2024): Commercial office lease rental rates reflect H1 2024 JLL and Colliers benchmarks. Real-time market changes in 2025/2026 require dynamic index updates.

2. Undisclosed Private Headcounts: Private multinational captives (e.g. Nvidia Pune, Alliant Vizag) do not publish isolated city headcounts. Kept as 'Data unavailable' rather than guessing.

3. Local-Only Deployment: Prototype currently operates strictly on local development environments; no public cloud deployment is configured.

NEXT STEPS & PRODUCT ROADMAP

1. Phase 1: Live ATS Integrations: Connect directly to Greenhouse, Lever, and Workday APIs to replace industry conversion benchmarks with live company historical pass rates.

2. Phase 2: Micro-Market Commercial Real Estate: Integrate live CRE data feeds to provide neighborhood-level lease rates (e.g., Whitefield vs ORR in Bengaluru; Madhapur vs Gachibowli in Hyderabad).

3. Phase 3: Expanded City Coverage: Expand data collection to 6 additional emerging hubs: Ahmedabad, Jaipur, Kochi, Indore, Vadodara, and Chandigarh.